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How can Publishers Build the Optimal AdStack in 2025?

These days publishers use their adtech stacks not only to sell ad space but to maximize the revenue generated from that space using the right tools. But first, what does the optimal adstack entail?

Navigating the intricate world of ad tech solutions can seem overwhelming for publishers aiming to boost ad revenue, and that is why having an optimized ad stack is so important.

However, setting up a programmatic ad stack is a challenge that both the publisher's developer, product, programmatic, and revenue operation teams suffer due to the expertise, time, and financial resources it demands.

So where do we start? To help you succeed as a publisher, let's explore what an ad tech stack comprises, its key components, and the **best practices for creating an optimal ad stack in 2025**.

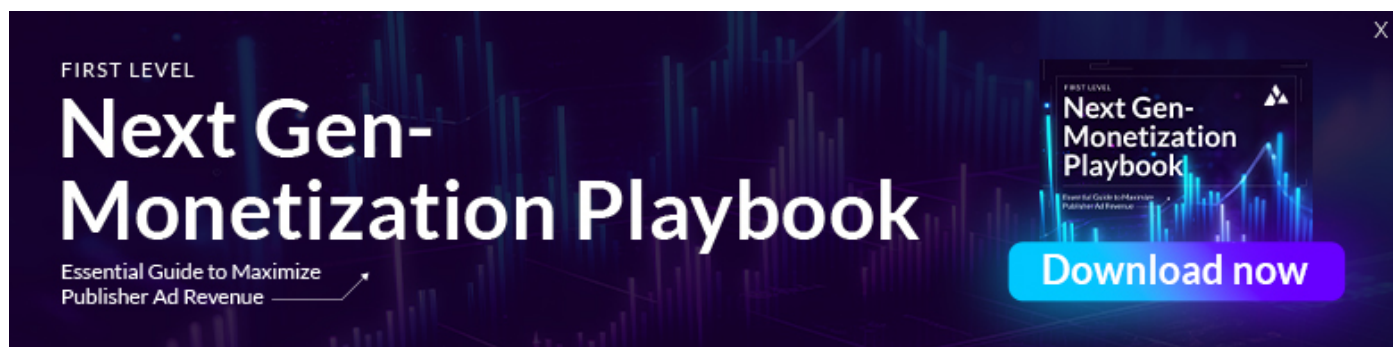
What is an ad stack in the publisher context?

An ad stack, often referred to as an **adtech stack**, is the **collection of technologies, systems, platforms, and tools that publishers use to manage** and deliver advertisements on their digital properties. This stack includes a combination of various ad tech components, each serving a specific purpose in the ad ecosystem.

Publishers adopt different adtech stack tools not only to sell ad space but also to maximize the revenue generated from that space. However, managing several revenue streams, consolidating, analyzing, troubleshooting, testing, and scaling changes is an endless nightmare.

Although there are [header bidding wrappers](#) that can combine vendors in one place, and scale publishers' ad operations as programmatic teams constantly search for different ways to monetize and increase revenue while improving user experience.

Learn more on how to help your developer, product, programmatic, and revenue operation teams seamlessly monetize your website with a single JavaScript on the page! Download the AY Next-Gen Monetization Playbook for Publishers.

A promotional banner for the 'Next Gen-Monetization Playbook'. The background is dark purple with a faint, glowing line graph. On the left, the text 'FIRST LEVEL' is in small white capital letters, followed by 'Next Gen-Monetization Playbook' in large, bold white font. Below this, in smaller white text, is 'Essential Guide to Maximize Publisher Ad Revenue'. On the right, there is a smaller image of the playbook cover, which also features the title and a line graph. Below this image is a bright blue button with the text 'Download now' in white. A small 'X' icon is in the top right corner of the banner.

The main components of a publisher ad stack

To create an effective ad stack, publishers must understand the main components that make up their technology arsenal. These components include

SSPs (Supply-Side Platforms)

An [SSP](#) is a gateway through which publishers manage and sell their ad inventory to advertisers. It plays a crucial role in optimizing ad revenue by connecting publishers with demand-side platforms (DSPs) and ad exchanges. SSPs normally offer several programmatic ad tools for ad sales optimization, such as [header bidding](#), and various forms of automated real-time ad auctions.

Data Management Platforms (DMPs)

Data Management Platforms (DMPs) are indispensable components of a publisher's ad tech stack since programmatic aggregates tons of data from multiple sources.

They are responsible for the collection, storage, and analysis of substantial volumes of audience data, which is leveraged programmatically to refine audience targeting and optimize ad campaigns for publishers. Recognized DMPs in the industry include [Adobe Audience Manager](#), [Oracle BlueKai](#), and [Nielsen DMP](#).

Ad networks

Ad networks acquire ad inventory from various publishers and consolidate it within a unified platform. Subsequently, they resell this ad space to advertisers and earn a commission on the sales. Ad networks simplify the process for publishers by eliminating the need to negotiate deals with each advertiser individually. Some well-known Ad networks include [Amazon Publisher Services](#), [Media.net](#), and [many more](#)

Ad Exchanges

An ad exchange is similar to an ad network, except it doesn't purchase the ad space. Instead, it establishes an open marketplace where advertisers and publishers can connect to buy and sell ad space and advertisements. Unlike Ad networks, they typically provide access to more inventory from more sources. [Google Ad Exchange](#) (AKA, The DoubleClick Ad Exchange), [Index Exchange](#), etc, are well-known ad exchanges.

How to build the optimal ad tech stack for publishers

Creating an optimal ad stack is essential for publishers looking to succeed in the programmatic advertising landscape. However, we know that it's a lot to take in, and that's why we segregate the components you need in your ad tech stack at every stage of your publishing journey.

While most beginners can get by only with a basic SSP and default analytics integration, medium-sized to large publishers need a more robust ad stack.

Medium-sized publishers: why go for a full ad stack?

When you reach a point where you've gotten a healthy-sized audience who accounts for a significant chunk of your revenue and can confidently label yourself as a medium-sized publisher or even bigger, it's time to explore more intricate aspects of your ad tech stack and ad operations.

This includes delving into advanced features like [header bidding solutions](#) and consent management tools (although, for GDPR or CCPA areas, it doesn't matter what size publisher you are, you must have this)

By incorporating header bidding, you might find it beneficial to transition to a more specialized ad server configuration or opt for a well-established full [ad stack solution that](#) can manage multiple revenue streams, analyze, unify, troubleshoot, test, deploy and revert changes by combining more than 11 vendors in one place to scale ad operations, monetize and boost revenue and UX.

Advanced Publishers

As you get to an advanced level, your users' data becomes a valuable asset in its own right. At this point, you enter the stage where you incorporate [a total ad revenue management tool](#) to grow revenue by gaining agility and scaling your resources.

Additionally, this is the phase in which you expand your pool of demand sources to elevate your CPM (cost-per-mile) rates significantly. Simultaneously, you optimize for mobile by establishing a dedicated mobile ad stack to cater to the increasing mobile audience.

Building your full adtech stack: what to consider

Building an optimal ad stack could present a challenge to many publishers, and not one you can solve overnight. This doesn't imply that a comprehensive, [all-in-one solutions](#) for publisher ad tech stack

Instead, it merely suggests that, in certain instances, you may need to attain a certain level of development before esteemed partners consider it mutually beneficial to collaborate with you.

Although, according to the [Nex-Gen monetization playbook](#), building an optimal ad stack is now made easy for the various publishers' teams to seamlessly monetize their websites using a single JavaScript on the page.

Header bidding

Discussing [header bidding](#) within the publisher's adtech stack comprises a few key elements. There's the underlying technology, typically a snippet of code in your website's header. **Then, you have the header bidding wrapper, serving as an interface to manage auction rules and demand partners.**

Noteworthy header bidding tools include [Amazon](#) (featuring Unified Ad Marketplace and Transparent Ad Marketplace), and [Prebid](#) (an open-source community). These tools streamline the integration of header bidding on your website by unifying the different components of the technology.

However, when it comes to mobile in-app publishers, the In-app header bidding is used. This is a feature commonly integrated into comprehensive Software Development Kit (SDKs) and is essential for maximizing CPMs in your in-app advertising strategy.

The key distinction between web and mobile header bidding is that the in-app header bidding process occurs entirely within your app's SDK, eliminating the need for an external wrapper. It's worth considering in-app header bidding when you've developed a substantial app with a growing audience, as medium and large publishers can reap significant rewards from its implementation.

Ad server

An ad server is a crucial tool that stores, manages, and delivers ads to websites and apps. It handles ad targeting, tracking, and reporting, ensuring the right ads reach the right audience at the right time.

They serve as a vital link in the programmatic chain, ensuring that, once an auction concludes, they deliver the appropriate message and choose the fitting ad for the specific ad slot.

Ad servers also make real-time automated decisions on which ads to show based on various factors and collect data on **metrics like video completion rates, ad impressions, clicks, viewability, and other key metrics**. Large publishers may need to explore advanced ad server options such as [OpenX](#), [Revive Adserver](#), [Keve](#), and more as they grow.

Video player

Video players are vital for **showcasing video ads in digital content like website videos or mobile apps**. For publishers looking to run video ads, it's not just about having a video player; it should also support running video ads and accepting winning bids from video header bidding like [Google Ad Manager](#).

Although all video players nowadays handle ad serving from GAM line items as well as their own auctioned ads. [Brightcove](#), [DaCast](#), [JW Player](#), etc, are just a few examples you can check out.

According to [Statistica's global video ad spend prediction](#), the industry is expected to grow annually at a rate of 6.80% from 2023 to 2027, with an anticipated market size of US\$229.80 billion by 2027. So, you may want to consider investing in an Online Video Player (OVP) that empowers you to effortlessly manage, monetize, and integrate video content with video header bidding ecosystems.

Brand safety and ad verification

Brand safety and ad verification tools are essential in the digital advertising landscape to counteract the prevalence of fraud and undesirable content. Advertisers seek assurances that their ads appear alongside safe content and are viewed by real human users, not automated bots or fraudulent traffic.

Ad tech experts have responded to this demand by developing tools such as Integral [Ads Science \(IAS\)](#), and others to ensure brand safety and verify the authenticity of ad views.

While automated solutions may not detect all instances of ad fraud or complex content issues, brand safety, and ad verification tools help mitigate common problems related to brand safety and traffic fraud.

Consent Management Platform (CMP)

Publishers must adhere to diverse online privacy and tracking regulations, like the [General Data Protection Regulation \(GDPR\)](#) in Europe, the [California Consumer Privacy Act \(CCPA\)](#) in California, and similar laws that apply to different countries and states.

A reliable Consent Management Platform (CMP) ensures compliance with these laws and enforces their mandates on your website. While some customization is possible, CMPs are structured to prioritize legal compliance.

Consider trying [InMobi's Consent Management Platform](#), as it gained prominence and achieved the highest market share in Q3'23, as reported in the [AY Q3 Industry Insights Report](#).

On the other hand, the primary purpose of an app Consent Management Platform (CMP) is to ensure that your app adheres to applicable privacy laws.

Typically, this involves a straightforward process, such as presenting users with a banner message upon app launch, offering them the option to opt out of tracking, and thus maintaining compliance with these regulations.

Identity solution

[Identity resolution technology](#) is crucial for both publishers and advertisers, to facilitate cross-device and cross-platform user identification, leading to a more holistic understanding of user behavior for targeted ads personalization.

Most identity solutions rely on first-party data, sourcing it from a network of publishers or other channels to generate valuable audience segments, akin to third-party data's audience segment creation.

If your data-gathering and audience segment strategies to increase your CPMs depend on third-party cookies, it's imperative to explore identity solutions and [alternatives to Google's third-party cookies termination](#) in 2024.

DMPs

Data Management Platforms (DMPs) are extensive data storehouses containing information about users, comprising first, second, or third-party data, although the use of third-party data has declined due to Google's decision to [phase out third-party cookies](#).

DMPs resolve the challenge of handling vast data volumes. For publishers, efficiently managing their user data to enhance revenue can be daunting. DMPs facilitate the combination of their data with data from other publishers, amplifying its effectiveness and value.

[Top-tier DMPs](#) offer real-time AI-driven insights, data from diverse sources beyond a publisher's user base, automated reporting, and data predictions. They include a management feature, allowing publishers to access, categorize, and integrate data from multiple sources into a single source to boost CPMs.

How to boost your publisher ad stack in 2025

In 2025, publishers need to be more aggressive in optimizing their ad stacks. In our experience, even ad stacks maintained and optimized for years may miss some of the best practices.

That's why **forward-thinking publishers use [a powerful wrapper](#) that lets them monetize their website with one single JavaScript on the page.** It combines more than 11 solutions, such as a tag manager, no-dev A/B test capabilities, namespaces, variables, the AY Data's power, and a suitable UX with the ability to implement, preview, and deploy changes all at once.

With capabilities like no code devs, faster integrations, instant feedback, time travel-like track history, lightning-fast testing, optimization at its finest, and holistic analysis, level up your publishing game with Assertive Yield.

Curating the optimal adstack doesn't have to be complicated, download the [Next-Gen Monetization Playbook for Publisher](#) to discover new ways to empower your developer, product, programmatic, and revenue operation teams to build the perfect adstack.

Say hello to the new era of publishing – it's AY-mazing!

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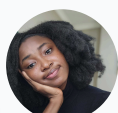
Ad Revenue Management

Check out the associated AY solution

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AY Prebid Server

Written by



Angela A. Oladipupo

Content Analyst

Angela is a wordsmith who skillfully transforms complex concepts into engaging content, making significant contributions to the advertising industry. Leveraging her wealth of expertise as a content writer, she crafts compelling narratives that seamlessly bridge technology and communication, whether with her pen or keyboard.

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